In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 264**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i). Section-A (01 – 10) contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 2 marks** for wrong answer.

(ii). Section – B (01 – 02) contains 2 Match the following Type questions. Each question having 4 statements in Column I & 5 statements in Column II with any given statement in Column I having correct matching with 1 or more statement (s) given in Column II. Each statement carries **+2 marks** for correct answer and **– 1 mark** for wrong answer.

(iii). Section-C (01 – 08) contains 8 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

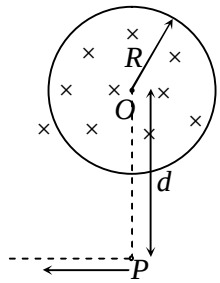
PART – I

SECTION – A
(One or More Than One Options Correct Type)

This section contains **10 multiple choice type questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

1. A 10 m long horizontal stainless steel wire AB of mass 1 kg whose end A is fixed, is connected to a massless string BC passing over a smooth pulley. String BC is connected to a container of mass 2 kg at end C. Water (density = 1×10^3 kg/m^3) is poured in the container at a constant rate of 2.25 litre/sec at t = 0. Also at t = 0, a pulse is generated at end A. Choose from the following the correct option(s).
   (A) Time taken by the pulse to reach point B is 0.612 sec
   (B) Time taken by the pulse to reach point B is 0.212 sec
   (C) Tension in the string at this moment is 33.77 N
   (D) Tension in the string at this moment is 24.77 N

2. In a cylindrical region of radius R, there exists a time varying magnetic field B such that $\frac{dB}{dt} = k(>0)$. A charged particle having charge q is placed at the point P at a distance d (> R) from its centre O. Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.



   (A) Work done by external agent is $\frac{q\pi R^2}{4}k$ if d = 2R
   (B) Work done by external agent is $\frac{q\pi R^2}{8}k$ if d = 4R
   (C) Work done by external agent is $\frac{q\pi R^2}{4}k$ if d = 4R
   (D) Work done by external agent is $\frac{q\pi R^2}{4}k$ if d = 6R

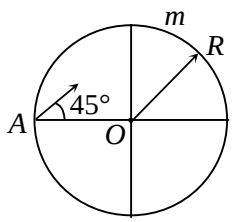
Space for Rough work

3. If the dimensions of length are expressed as $G^x c^y h^z$; where G, c and h are the universal gravitational constant, speed of light and Planck's constant respectively, then

(A) $x = \frac{1}{2}, y = \frac{1}{2}$ (B) $x = \frac{1}{2}, z = \frac{1}{2}$

(C) $y = \frac{1}{2}, z = \frac{3}{2}$ (D) $y = -\frac{3}{2}, z = \frac{1}{2}$

4. A ring of mass m and radius R is placed on a frictionless horizontal surface. A particle of mass m is projected from point A with velocity v at an angle of 45° with AO as shown. The correct statements are



(A) The particle reaches the same point A on the ring after time $\frac{4R\sqrt{2}}{v}$.

(B) Magnitude of impulse transformed during first collision is $\frac{mv}{\sqrt{2}}$.

(C) Magnitude of impulse transformed during second collision is $\frac{mv}{\sqrt{2}}$.

(D) Particle reaches diametrically opposite point on the ring in time $\frac{2R}{v}$.

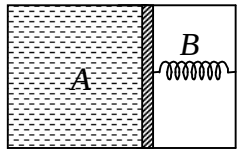
5. Two projectile are thrown at the same time from two different points. The projectile thrown from the origin has initial velocity $3\hat{i} + 3\hat{j}$ with respect to earth. The projectile has initial velocity $a\hat{i} + b\hat{j}$ with respect to earth thrown from the point (10, 5). ($\hat{i}$ is a unit vector along horizontal, $\hat{j}$ along vertical). If the projectile collides after two second, then the

(A) value of a is -2 (B) value of a is $\frac{1}{2}$

(C) value of b is $\frac{1}{2}$ (D) value of b is –2

Space for Rough work

6. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant k is connected with piston and the wall of the container as shown. Initially spring is unstretched. Gas in chamber A is allowed to expand. Let in equilibrium spring is compressed by x_0. Then



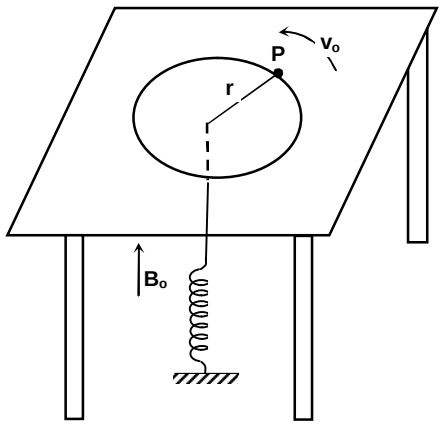
(A) final pressure of the gas is $\frac{kx_0}{S}$

(B) work done by the gas is $\frac{1}{2}kx_0^2$

(C) change in internal energy of the gas is $\frac{1}{2}kx_0^2$

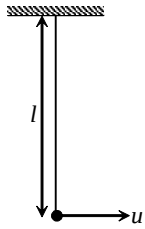
(D) temperature of the gas is decreased

7. A particle mass m charge q is moving on a circular path on the surface of a frictionless table with speed v_o where a magnetic field B_o exists uniformly over the whole region. It is attached by a string which passes through a hole in the table to a spring as shown.
The spring is stretched by x_o. If now the magnetic field is increased slowly to $2B_o$, which of these are incorrect?
(A) The extension in the spring will increase
(B) The speed of the particle will increase
(C) The speed of the particle will decrease
(D) The kinetic energy of the particle will decrease



Space for Rough work

8. A particle of mass m is attached to one end of the light inextensible string and other end of the string is fixed in vertical plane as shown. Now particle is given horizontal velocity $u = \sqrt{\frac{5}{2}gl}$, then which of the following statement/s is/are correct.



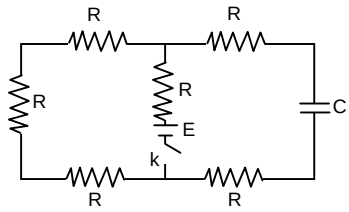
(A) The string makes angle with the downward vertical is 60^0 when acceleration of the particle is horizontal.

(B) The tension in string is 2 mg when acceleration of the particle is horizontal.

(C) The speed of the particle is $\sqrt{\frac{3}{2}gl}$ when acceleration of the particle is horizontal

(D) The particle will not complete the circle.

9. In the given circuit initially the capacitor is uncharged. At t = 0 the key k is closed. Choose the correct option(s)



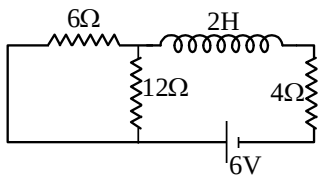
(A) The time constant of the circuit is $\frac{11RC}{4}$.

(B) The time constant of the circuit is $\frac{11RC}{5}$.

(C) Current through the capacitor after one time constant is $\frac{3E}{11R}\left(\frac{1}{e}\right)$.

(D) Current through the capacitor after one time constant is $\frac{5E}{11R}\left(\frac{1}{e}\right)$.

10. For the circuit shown in the figure, which of the following statements is/are correct



(A) Its time constant is 0.25 second

(B) In steady sate, current through inductance will be equal to zero.

(C) in steady state, current through the battery will be equal to 0.75 amp.

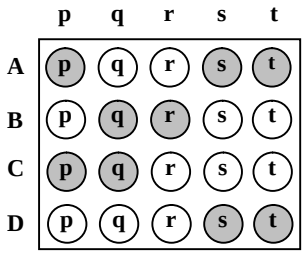
(D) none of these

Space for Rough work

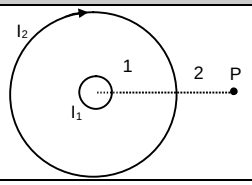
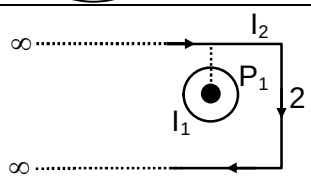
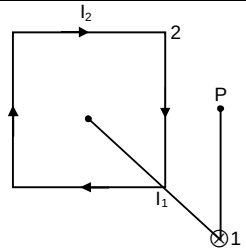
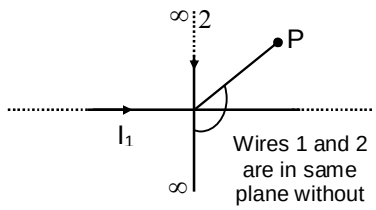
SECTION – B

Matrix-Match Type

*This section contains 2 questions. Each question contains statements given in two columns which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A-p, s and t; B-q and r; C-p and q; and D-s and t; then the correct darkening of bubbles will look like the following.

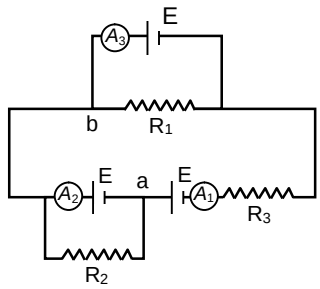


1. Match the Column:

Column – I		Column – II	
(A)		(p)	1 and 2 attract each other
(B)		(q)	1 and 2 repel each other
(C)		(r)	1 and 2 do not exert any force on each other
(D)		(s)	Magnetic field at point P due to wires 1 and 2 are perpendicular to each other.
		(t)	Magnetic field at point P due to wires 1 and 2 are in same direction.

Space for Rough work

2. In the circuit shown in figure, $R_1 = R_2 = R_3 = 3\Omega$ and e.m.f. of each cell is E = 4V and negligible internal resistance. All ammeters are ideal. Match the following:



Column - I		Column - II	
(A)	Reading of ammeter A_1 in ampere is	(p)	4/3
(B)	Reading of ammeter A_2 in ampere is	(q)	8/3
(C)	Reading of ammeter A_3 in ampere is	(r)	4
(D)	Potential difference between point a and point b in volt is	(s) (t)	zero 2

SECTION – C

(One Integer Value Correct Type)

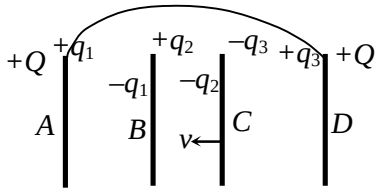
This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A point source of power 4W is placed 1m below the free surface of liquid whose refractive index is '$\frac{2}{\sqrt{3}}$'. Find the rate of transfer of energy (in watt) from the liquid surface to air. Ignore any absorption or scattering of light energy.

2. A double convex lens forms a real image of an object on a screen which is fixed. Now the lens is given a constant velocity v_0 along its axis and away from the screen. For the purpose of forming the image always on the screen, the object is also required to be given an appropriate velocity. Find the velocity of the object at the instant its size is n times the size of image. (Take n = 1/2 and v_0 = 1 m/s)

Space for Rough work

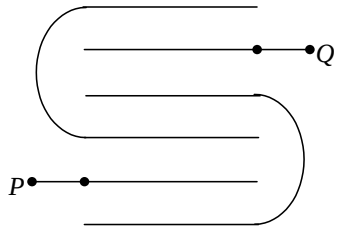
3. In the diagram there are four conducting plates A, B, C and D placed parallel to each other at equal separation L. If plate C starts moving towards plate B with velocity v. Find the current (in mA) flowing in the wire connecting A and D. (assume all other plates to be fixed) (Given: q_2 = 5μC, q_3 = 10μC, v = 10 m/s, L = 0.05 m)



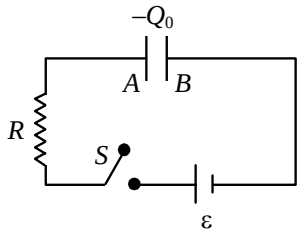
4. A thin uniform rod of mass 'm', length 'ℓ' rotates uniformly about a vertical axis ω, form a conical pendulum, the upper end of the rod is hinged and θ is angle between rod and vertical.

If the change of angular momentum of the rod about hinge is $\frac{m\ell^2}{x}\omega^2 \sin 2\theta$ then the value of 'x' will be

5. Identical metal plates are located in air at equal distances *d* from one another. The area of each plate is equal to A. The capacity of system between points *P* and *Q* is $\frac{K\varepsilon_0 A}{d}$. Then the value of 'K' is



6. The figure shows a *RC* circuit with a parallel plate capacitor. Before switching on the circuit, plate *A* of the capacitor has a charge $-Q_0$ while plate *B* has no net charge. Now, at *t* = 0, the circuit is switched on. How much time (in second) will elapse before the net charge on plate *A* becomes zero. (Given *C* = 1μF, Q_0 = 1mC, ε = 1000 V and $R = \frac{2\times 10^6}{\ln 3}\Omega$)



7. A ring of mass 3 kg is rolling without slipping with linear velocity 1 m/sec on a smooth horizontal surface. A rod of same mass is fitted along its one diameter. Find total kinetic energy of the system (in J).

8. Potential energy of a particle moving along x-axis is given by U = $\frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state.

Space for Rough work

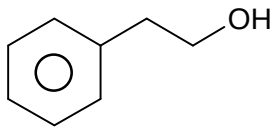
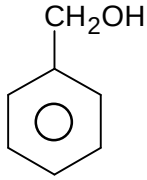
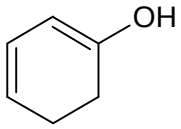
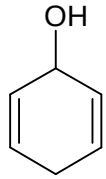
Chemistry

PART – II

SECTION – A
Straight Objective Type

This section contains **10 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

1. Which of the following compound produce any gas when reacts with water?
   (A) KO_2 (B) CaC_2
   (C) Mg_3N_2 (D) $Ba(NO_3)_2$

2. Which of the following compound does exhibit mesomeric effect?
   (A) 
   (B) 
   (C) 
   (D) 

3. Which of the following compound of carbon forms acidic solution when reacts with water at room temperature?
   (A) CO (B) $COCl_2$
   (C) CCl_4 (D) CH_4

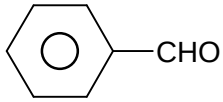
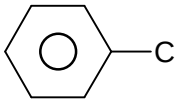
4. Which of the following element is present in silicones?
   (A) Carbon (B) Hydrogen
   (C) Oxygen (D) Sulphur

5. The boiling point of which of the following aqueous solution(s) is/are greater than 100^oC?
   (A) 0.2 M NaCl (B) 0.1 M $AlCl_3$
   (C) 1 M KNO_3 (D) 2 M $BeSO_4$

Space for Rough work

6. Which of the following is/are the coordination isomer(s) of $[Co(NH_3)_6]\ [Cr(CN)_6]$?
(A) $[Cr(NH_3)_6]\ [Co(CN)_6]$ (B) $[Cr(NH_3)_4(CN)_2]\ [Co(CN)_4(NH_3)_2]$
(C) $[Cr(NH_3)_3(CN)_3]\ [Co(NH_3)_3(CN)_3]$ (D) $[Cr(NH_3)_4]\ [Co(CN)_4]$

7. Which of the following two compound(s) form different number of gases on heating?
(A) $LiNO_3$ and $NaNO_3$ (B) $MgCl_2.6H_2O$ and $Na_2SO_4.10\ H_2O$
(C) $FeSO_4$ and $Fe_2(SO_4)_3$ (D) $CaCO_3$ and $BeCO_3$

8. Which of the following compounds can be distinguished by Fehling's test?
(A) CH_3CHO and CH_3COCH_3 (B) CH_3CH_2CHO and 
(C)  and CH_3COCH_3 (D) CH_3CHO and HCHO

9. The hcp and ccp structure for a given element would be expected to have
(A) the same coordination number (B) the same density
(C) the same packing fraction (D) the same edge length

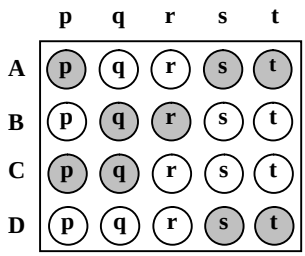
10. In which of the following pair, the dipole moment of first compound is greater than that of the second compound?
(A) NH_3, NF_3 (B) $CHCl_3$, CCl_4
(C) CO_2, SO_2 (D) BF_3, NF_3

Space for Rough work

SECTION – B

Matrix-Match Type

*This section contains 2 questions. Each question contains statements given in two columns which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A-p, s and t; B-q and r; C-p and q; and D-s and t; then the correct darkening of bubbles will look like the following.



1. Match the radicals mentioned in Column-I with their reactions mentioned in Column-II.

Column – I		Column – II	
(A)	Fe^{2+}	(p)	Forms precipitate with KCN, which is soluble in excess of the reagent.
(B)	Hg^{2+}	(q)	Forms precipitate with NaOH and NH_4OH, which is insoluble in both excess of reagent.
(C)	Pb^{2+}	(r)	forms coloured precipitate with KI, which is soluble in excess of reagent.
(D)	Ag^{+}	(s)	forms black precipitate with H_2S which is soluble in hot and dilute HNO_3.

2. Match the aqueous solutions of acids, bases and salts mentioned in Column-I with their pH and nature mentioned in Column-II.

Column – I		Column – II	
(A)	HCOOH (p^{K_a} = 3.74, 0.1 M) + HCOONa (0.1 M)	(p)	Acidic buffer at its maximum capacity
(B)	CH_3COOH (0.1 M) + HCl (0.1 M)	(q)	behaves as a buffer
(C)	CH_3COOH (p^{K_a} = 4.74, 0.1 M) + NH_4OH (p^{K_a} = 9.26, 0.1 M)	(r)	pH < 7 at 298 K
(D)	HCOONa (300 mL of 0.1 M) + HCl (100 mL of 0.1 M)	(s)	pH = 7 at 298 K

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. 4.48 litre of an ideal gas at STP requires 12 calorie to raise the temperature by 15°C at constant volume. What is the value of C_P of the gas in calorie unit?

2. How many maximum number of nitro product(s) is/are formed when $CH_3CH_2CH_3$ reacts with fuming HNO_3?

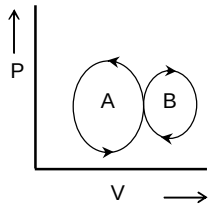
3. The half-life of a zero order reaction is 4 sec. How much seconds is needed for 75% completion of the reaction?

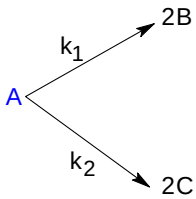
4. What is the sum of the number of equivalents of glucose and the number of equivalents of phenylhydrazine required to form one equivalent of glucosazone?

5. Compound X dissociates according to the reaction $2X(g) \rightleftharpoons 2Y(g) + Z(g)$, with degree of dissociation α which is small compared with unity, if expression for α in terms of equilibrium constant K_p and total pressure P is given as $\alpha = \left(\frac{2K_p}{P}\right)^{1/n}$. The value of n is

Space for Rough work

6. In the present graphs, the area of circle A and B are 25 unit and 20 unit respectively. Work done will be ____ unit.



7. For a hypothetical elementary reaction, 

Where $\frac{k_1}{k_2}=\frac{1}{2}$ initially only 2 moles of A are present. Total number of moles of A,B and C at the end of 75% reaction are X. What will be the value of 2X

8. A vessel of 250 litres was filled with 0.01 mole of Sb_2S_3 and 0.01mole of H_2 to attain the equilibrium at 440^0C as $Sb_2S_3(s)+3H_2(g)\rightleftharpoons 2Sb(s)+2H_2S(g)$. After equilibrium, the H_2S formed was analysed by dissolved it in H_2O and treating with excess of Pb^{2+} to give 1.19g of PbS $(M_{Pb}=207)$ as precipitate. The value of K_c at 440^0C is _____.

Space for Rough work

Mathematics

PART – III

SECTION – A

Multiple Correct Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

1. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a}\times\left(\vec{a}\times\vec{c}\right)+\vec{b}=0,$ then

   (A) $\vec{b}$ is in the plane of $\vec{a}$ and $\vec{c}$ (B) $\vec{b}$ is parallel to $\vec{a}\times\vec{c}$

   (C) angle between $\vec{a}$ and $\vec{c}$ is 30° (D) angle between $\vec{a}$ and $\vec{c}$ is 150°

2. The number of solutions of equation $8\sin x=\frac{\sqrt{3}}{\cos x}+\frac{1}{\sin x}$

   (A) 6 if $x\in(0,2\pi)$ (B) 4 if $x\in(0,\pi)$

   (C) 5, if $x\in\left(0,\frac{3\pi}{2}\right)$ (D) There does not exist any solution

3. If A and B are two orthogonal matrices of order n and $\det(A)+\det(B)=0,$ then which of the following must be correct

   (A) $\det(A+B)=\det(A)+\det(B)$ (B) $\det(A+B)=0$

   (C) A + B = 0 (D) None of above are always correct

Space for Rough work

4. Consider the function g defined by $g(x)=\begin{cases} x^2\sin\frac{\pi}{x}+(x-1)^2\sin\left(\frac{\pi}{x-1}\right), & x\neq 0,1 \\ 0, & \text{if } x=0,1 \end{cases}$ then which of the following statement (s) is/are correct?

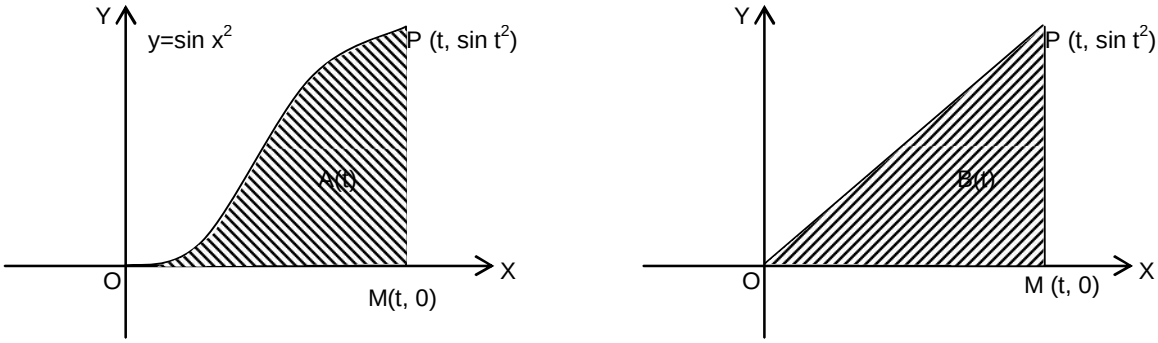
(A) $g(x)$ is differentiable $\forall x\in R$

(B) $g'(x)$ is discontinuous at $x=0$ but continuous at x = 1

(C) $g'(x)$ is discontinuous at both $x=0$ and $x=1$

(D) Rolle's theorem is applicable for $g(x)$ in [0, 1]

5. The figure shows two regions in the first quadrant.



A(t) is the area under the curve , $y=\sin x^2$ from 0 to t and B (t) is the area of the triangle with vertices O, P and M (t, 0).Then $\lim\limits_{t\to 0}\frac{A(t)}{B(t)}$ equals

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$

(C) $\frac{1}{2}$ (D) 1

6. Diagonals of a square are along pair of lines $2x^2-3xy-2y^2=0$. If (2, 1) is a vertex of square then sides of square can be

(A) $3x-y+5=0$ (B) $x+3y+5=0$

(C) $2x-y-5=0$ (D) $x-2y=0$

Space for Rough work

7. If three planes $P_1 \equiv 2x + y + z - 1 = 0$, $P_2 \equiv x - y + z - 2 = 0$ and $P_3 = \alpha x - y + 3z - 5 = 0$ intersects each other at point P on XOY plane and at point Q on YOZ plane, where O is the origin then identify correct statement(s)
(A) The value of α is 4
(B) Straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$.
(C) The length of projection of $\overrightarrow{PQ}$ on x – axis is 1.
(D) Centroid of the triangle OPQ is $\left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$

8. In the expansion of $(2x - 3y + z)^{10}$, then which of the following statement(s) is/are correct?
(A) The expansion will contain 66 terms
(B) Sum of all the coefficients is 0.
(C) The coefficient of $x^2y^3z^5$ equals $\left(\frac{-10! \times 9}{5!}\right)$
(D) None of these

9. The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} {}^{100}C_k (x-2)^{100-k} 3^k$ is also equal to
(A) number of ways in which 50 identical books can be distributed in 100 students, if each student can get atmost one book.
(B) number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together.
(C) numbers of dissimilar terms in $(x_1 + x_2 + x_3 + ... + x_{50})^{51}$.
(D) $\frac{2.6.10.14.....198}{50!}$

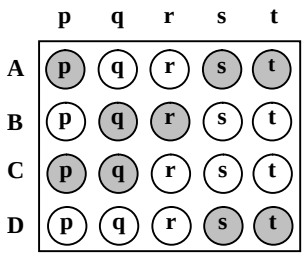
10. Let the complex number $z = x + iy$ (where $x, y \in R$) satisfy $\cot^{-1}(\log_3 |2z+1|) > \cot^{-1}(\log_3 |2z-1|)$, then x can be
(A) –1 (B) –2
(C) 1 (D) 2

Space for Rough work

SECTION – B

Matrix-Match Type

*This section contains 2 questions. Each question contains statements given in two columns which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A-p, s and t; B-q and r; C-p and q; and D-s and t; then the correct darkening of bubbles will look like the following.



1. Match the columns:

	Column - I		Column - II
(A)	The maximum value of $\left(\cos(2A+\theta)+\cos(2B+\theta)\right)$, where A, B are constant, is	(p)	$2\sin(A+B)$
(B)	The maximum value of $(\cos 2A+\cos 2B)$, where (A + B) is constant and $A,B\in\left(0,\frac{\pi}{2}\right)$, is	(q)	$2\sec(A+B)$
(C)	The minimum value of $(\sec 2A+\sec 2B)$, where (A + B) is constant and $A,B\in\left(0,\frac{\pi}{4}\right)$, is	(r)	$2\cos(A+B)$
(D)	The minimum value of $\sqrt{(\tan\theta+\cot\theta-2\cos(A+B))}$ where A, B are constants and $\theta\in\left(0,\frac{\pi}{2}\right)$, is	(s)	$\lvert 2\cos(A-B)\rvert$
		(t)	$2\cos(A-B)$

Space for Rough work

2. A plane π passes through the point (1,1, 1) and is parallel to the vectors $\vec{a}$ = (–1, 1, 0) and $\vec{b} = (1,0,-1)$ π meets the axis in A, B and C forming the tetrahedron OABC where O is origin.

	Column - I		Column - II
(A)	Volume of the tetrahedron OABC in cubic units	(p)	$\frac{9}{2}$
(B)	Area of the faces of the tetrahedron in square units	(q)	$\frac{9\sqrt{3}}{2}$
(C)	Distance of the points $\left(3\sqrt{3}/2, 3\sqrt{3}, 3\right)$ form the plane π in units	(r)	$\sqrt{3}$
(D)	Distance of the faces of the tetrahedron from the opposite vertex	(s)	3

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. The roots of the equation $z^4 + az^3 + (12 + 9i)z^2 + bz = 0$ are the vertices of a square then area of square is _____

2. Let $\beta = \int_0^1 \frac{2 + 3x + 4x^2}{2\sqrt{x^2 + x + 1}}\, dx$ then find the value of β^2.

3. The triangle ABC, right angled at C, has median AD, BE and CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60 and area of $\Delta ABC = 100(k)^2$ then k is _____

Space for Rough work

4. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality k > 0 depends on the acceleration due to gravity and the geometry of the hole. If time to empty the tank be T minutes, then $\frac{T}{10} =$

5. If $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{\sqrt{2n+2}-\sqrt{2n-2}}{2+\sqrt{n^2-1}}\right) = \frac{\pi}{2} + \tan^{-1} k$ then find the value of k^2

6. If x, y, z are positive real numbers and $x + y + z = 1$, then find the minimum value of $\left(\frac{4}{x}+\frac{9}{y}+\frac{16}{z}\right).\frac{1}{27}$ is _____

7 Let a, b, c be the sides of a triangle ABC and Δ be its area. If $a = 2, \Delta = \sqrt{3}$ and $a\cos C + \sqrt{3}\, a \sin C - b - c = 0$. Then the semi perimeter of the ΔABC is _____

8. Let slope of sides AB and BC of a ΔABC are roots of the equation $x^2 - 4x + 1 = 0$, (ΔABC lying in the first quadrant and B is origin) and inradius of the triangle is 1 unit. If co – ordinates of its incentre are (a, b) then $\left(a^2 + b^4\right)$ is _____

Space for Rough work